

SC 600: Research Methodology

Course Code: SC 600

Course Name: Research Methodology

L-T-P-C: 1-0-0-1

Prerequisites: None

Intended for: M.Tech(R) and Ph.D

Mutual Exclusion: None

Approval: 65th BoA

Preamble

This course delves into the various research methodologies relevant to computing and electrical engineering. The focus is on understanding the research process and how research problems in computing, electronics, communication, electrical engineering, and related areas are formulated and addressed systematically. Students will learn how to identify research problems, conduct literature reviews, formulate research questions and hypotheses, and apply appropriate scientific and engineering methods to solve research problems. Students will become familiar with theoretical, experimental, and computational approaches to research, including mathematical modelling, simulations, experimental design, data analysis, and numerical methods. The course will also emphasize research ethics, scientific integrity, and best practices in technical and scientific communication. The course will culminate in developing skills for preparing research reports and technical documents using LaTeX, based on the students' own research work.

Course Modules with quantitative lecture hours

- **Research Methodology and Literature Review:** Fundamentals of research, research process and design, data collection and analysis, literature review, research gaps, and research contribution. (3 Lectures)
- **Tools for Computing, Data Analysis, and Scientific Documentation:** Programming, scientific computing, data analysis, computational platforms, and LaTeX for research. (3 Lectures)
- **Scientific Communication and Intellectual Property:** Research writing, publications, presentations, scientific visualization, and intellectual property protection. (4 Lectures)
- **Artificial Intelligence for Research:** Applications of generative AI in research, its limitations, biases, and responsible use. (2 Lectures)
- **Research Ethics and Integrity:** Ethical conduct, research integrity, plagiarism, publication ethics, data management, and responsible research practices. (1 Lecture)
- **Professional and Career Development:** Research career planning, professional development, networking, funding opportunities, research planning, and work-life balance. (1 Lecture)

Laboratory/practical/tutorial Modules: N.A.

Text Books

1. Soumitro Banerjee, *Research Methodology for Natural Sciences*, IISc Press, 2022
2. Wayne Booth, Gregory Colomb, Joseph Williams, Joseph Bizup, William Fitzgerald, *The Craft of Research*, 4th edition, The University of Chicago Press, 2016



References

N.A.